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EDUCATION

Indian Institute of Information Technology, Raichur
Bachelor of Technology in AI and DS

Sept 2023 – Expected May 2027

GPA: 8.05 / 10.0

- **Coursework:** Artificial Intelligence, Machine Learning, Deep Learning for Differential Equations, Agentic AI, Reinforcement Learning, Data Structures and Algorithms, Operating Systems, Database Management Systems.

RESEARCH EXPERIENCE

Indian Institute of Technology Hyderabad
Research Intern (Advisor: Dr. R. Maheswaran)

Jan 2026 – July 2026

Hyderabad, India

- Contributing to ML for hydrology and Earth observation across multiple projects under review/preprint (results not yet disclosable).
- Developed an LSTM Encoder-based learnable bottleneck for Extreme Precipitation Event modeling.
- Designed Physics-Informed Frequency-Decomposed Transformers for temporal downscaling of GRACE satellite data.
- Engineered a Residual U-Net CVAE for multi-channel geophysical data with custom spatial regularization and physics-informed learning.
- Core member of the research group's hackathon team, winning **1st Place at Riverathon 2026**.

PUBLICATIONS & PREPRINTS

Does Smarter Acquisition Beat Random Sampling? A Fully-Controlled Study of Submodular, Density-Aware Active Learning for Fine-Grained Classification *[Preprint]*

[DOI |](#)

- Built a multi-modal active learning pipeline for 10,000-class fine-grained classification (iNaturalist): batched submodular optimization (CELF), density-weighted outlier rejection, and multi-exit self-contrastive learning, backed by crash-resilient checkpointing.
- Ran a fully-controlled experiment (matched epochs, rounds, budget, seeds) showing engineered acquisition underperforms random sampling once compute is matched – a rigorous finding aligned with recent AL benchmarking critiques.
- Measured a **14–96×** peak GPU memory reduction for submodular selection vs. a dense $\mathcal{O}(N^2)$ baseline, and validated pseudo-labeling at **89–96%** accuracy against held-out ground truth via a leakage audit.

PROJECTS

Deep Learning Solutions for Biharmonic Equations

[Code](#)

Python, PyTorch, CUDA, PINNs, Deep Ritz Method

- Developed a comparative framework for solving fourth-order **Biharmonic PDEs** using **Physics-Informed Neural Networks (PINNs)** and the **Deep Ritz Method (DRM)**.
- Implemented both strong-form and variational PDE solvers using automatic differentiation and energy minimization.
- Achieved **1.4% relative error** with PINNs while demonstrating a **10×** GPU speedup using DRM.

Agentic Legal Assistant (Law-Agentic System)

[Code](#)

LangGraph, LangChain, FastAPI, Firebase, Google Gemini

- Built a multi-role **Agentic AI** legal assistant using **LangGraph** and a **RAG** pipeline over Indian legal codes.
- Deployed with FastAPI and Firebase for complaint filing, case assignment, and legal workflow automation.

Reinforcement Learning for Lunar Landing

[Code](#)

PyTorch, Gymnasium, Stable Baselines3

- Developed a PPO-based reinforcement learning agent for LunarLander-v3 using Stable Baselines3 with 16 parallel environments for accelerated and stable learning.

TECHNICAL SKILLS

Languages: Python, C++, C, Rust, SQL

Frameworks & ML: NumPy, Pandas, PyTorch, LangChain, LangGraph, PINNs, Deep Ritz Method, Django, FastAPI

Tools & Platforms: CUDA, Git, SQLite, PostgreSQL, Google Colab, Jupyter Notebook, VS Code

Profiles & Competitions: LeetCode (@[abhay-tiwari](#)): 250+ Solved | **Top 120 Rank** – Amazon ML Challenge 2025